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THE ROLE OF BEZAFIBRATE IN LOWERING RISK FACTORS FOR CARDIOVASCULAR DISEASE IN DIABETIC PATIENTS

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ABSTRACT

A number of hypolipidaemic drugs have been shown to lower the raised plasma fibrinogen concentrations indicating that lipid and haemostatic system may act synergistically in pathogenesis of cardiovascular diseases. The current study examines the possible role of lipid lowering drug bezafibrate to induce change in blood coagulation system and the architecture of fibrin network in addition to its known effects on lipid profile in diabetic patients. Methods based on turbidity for measurement of mass length ratio of fibrin fibers, permeability of the networks and their tensile strength have been used to assess the alterations induced by lipid lowering agent.

The study group consisted of 38 patients (18 male and 20 female; age 40 – 65 years). The patients were categorized into two groups. Group one included 20 diabetic patients, selected from Liaquat National Hospital, Karachi, with normal lipid profile as control (9 male, age 51.88 ± 1.93 years; 11 female, age 48.36 ± 2.44 years). Second group included 18 diabetic patients, selected from National Institute of Cardiovascular Disease, Karachi, with altered lipid profile i.e. hypertriglyceridaemia and/or hypercholesterolemia (9 male, age 47.33 ± 3.08 years; 9 female, age 48.66 ± 3.99 years). Blood samples were collected after 12-14 hours fasting at the start of the study and after administration of Bezafibrate (200 mg TDS) for 4 weeks.

The samples were analyzed for blood parameters like cholesterol, low-density lipoprotein-cholesterol (LDL-C), high-density lipoprotein-cholesterol (HDL-C), triacylglycerol (TG), and fibrin network characteristics. In addition to the changes in lipid profile, significant alterations were observed in the characteristics of fibrin networks by the drug. Amongst the changes induced are the properties of networks that make the networks more lysable by fibrinolytic agents. Clinical and pharmacological implications of these findings need further investigations.

Keywords: Bezafiberate, Fibrinogen, Cardiovascular Diseases.

INTRODUCTION

Diabetes is one of the most common metabolic disorders in the elderly and diabetic patients frequently develop cardiovascular disease (Kannel *et al.*, 1990). The increased risk of cardiovascular complications is associated with raised levels of plasma lipids, cholesterol and low-density lipoprotein-cholesterol (LDL-C). Elevated fibrinogen level has been shown to be an independent risk factor for cardiovascular disease and is almost comparable with other risk factors, such as, age, blood pressure, adiposity, cigarette smoking, LDL-cholesterol and diabetes (Meade *et al.*, 1986; Kannel *et al.*, 1987; McGill and Ardlie, 1990; Ganda and Arkin, 1992).

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Laboratory studies have indicated how fibrinogen affects several pathways that are known or are suspected to be involved in thrombogenesis, principally the degree of blood viscosity, the amount of fibrin produced when the coagulation process is initiated, clot deformability, platelet aggregability and atherogenesis (Meade, 1994). This evidence suggests that high fibrinogen levels are of causal significance. On the other hand, studies of polymorphisms associated with fibrinogen levels (Zito *et al.*, 1997; Doggen *et al.*, 2000) have been equivocal regarding the relationship of the former with clinical events of CHD.

A number of hypolipidaemic agents have been shown to lower the raised plasma fibrinogen concentration. These findings have enhanced the interest in identifying agents that can synergistically normalize the level of fibrinogen and lipid profile and also has the capability to induce alteration in fibrin network structure. The agents with this capacity are the fabric acid derivatives of gemfibrozil and bezafibrate (Lakatos *et al.*, 1996). Bezafibrate reduces the incidence of myocardial infarction (MI) in patients with metabolic syndrome (MS) during long-term follow-up (Tenenbaum *et al.*, 2005). It also delays the onset of type 2 diabetes in obese patients and in patients with coronary artery disease over a long-term follow-up period (Tenenbaum *et al.*, 2004; 2005). Bezafibrate has also been shown to cause some reduction in major events of CHD (The Bezafibrate Infarction Prevention Trial 1993; Ibid, 2000; Ericsson *et al.*, 1996).

The current study explores the role of fibrinogen in vascular complication in diabetic patients and its prevention through modification of fibrin network structure by the hypolipidaemic agent bezafibrate. This has enhanced to interest in identifying agents that can normalize elevated plasma fibrinogen and lipid levels

MATERIALS AND METHODS

The study group consisted of 38 patients (18 male and 20 female; age 40-65 years). The patients were categorized into two groups. Group one included 20 diabetic patients, selected from Liaquat National Hospital, Karachi, with normal lipid profile as control (9 male, age 51.88 ± 1.93 years; 11 female, age 48.36 ± 2.44 years). Second group included 18 diabetic patients, selected from National Institute of Cardiovascular Disease, Karachi, with altered lipid profile i.e. hypertri-glyceridaemia and/or hypercholе-sterolemia (9 male, age 47.33 ± 3.08 years; 9 female, age 48.66 ± 3.99 years). Age, sex, weight, blood pressure, duration of diabetes, duration of complication, type of diabetes and type of treatment were recorded. Family history, personal history and past medical history were also taken.

Blood samples were collected after 12-14 hours fasting at the start of the study and after administration of 200 mg TDS Bezafibrate (Lipocor, Afroze Chemical Industries Pvt. Ltd, Karachi) for 4 weeks.

Kits for determination of total cholesterol, TG, HDL-C were supplied by Lab System, kit for blood glucose was supplied by Biolabo and plasma fibrinogen was determined by Clauss' method (Clauss, 1957).

Plasma for fibrin Network structure Study:
Blood was collected 3.8% trisodim citrate in a ratio of 9:1. Thirty five µl Trasylol (10000 KIU/ml) was added in 10 ml of the above sample to inhibit fibrinolytic activity. The blood was then centrifuged at 2400xg for 10 minutes to obtain platelet poor plasma and then recentrifuged at 4000xg for 5 minutes to obtain platelet free plasma. The concentration of fibrinogen in each sample was determined by Clauss's method. Network fiber thickness (μ_T), calculated from turbidity, permeability or Darcy's constant (τ) and compaction were determined according to Nair *et al.* (1991). Permeability, μ_T, τ and compaction were measured from networks developed immediately after blood collection. To minimize experimental error, measurement of μ_T and compaction were made in duplicate. Measurement of plasma fibrinogen concentration was made in duplicate.

Statistical analyses were performed using the statistical software STATISTICA (StatSoft, USA). The significance was determined by employment two sampled Student's t-test.

RESULTS

Results are expressed as percent of the respective index in the control group. The values of the control indices taken as 100.

Figure 1 shows alterations in lipid profile and blood glucose by bezafibrate. When compared with the respective control index, cholesterol ($p<0.001$), triacylglycerol ($p<0.001$) and LDL-C ($p<0.05$) were significantly higher in patients, before admnistration of bezafibrate. Cholesterol ($p<0.001$), triacylglycerol ($p<0.001$) and glucose ($p<0.05$) levels decreased after adminstration of bezafibrate as compared to the levels before administration of the drug.

Figure 2 shows alterations in fibrin network characteristics following the

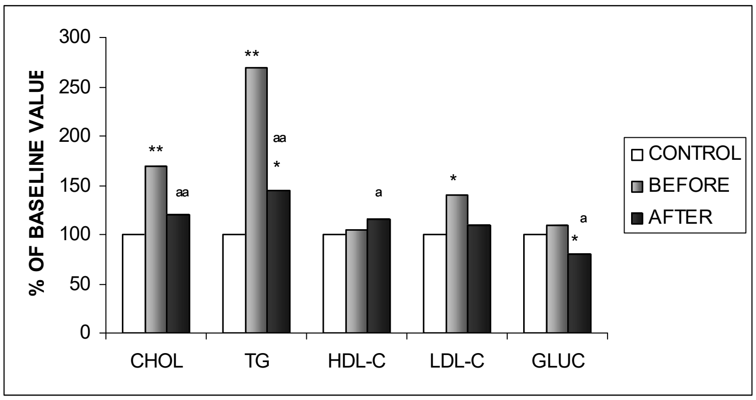

Fig. 1. Alteration in Lipid Profile and Blood Glucose by Bezafibrate.
*$p<0.05$, **$p<0.001$ when diabetic patients before and after treatment are compared with control group.
[a]$p< 0.05$, [aa]$p< 0.001$ when diabetic patients after treatment with bezafibrate are compared with before treatment.

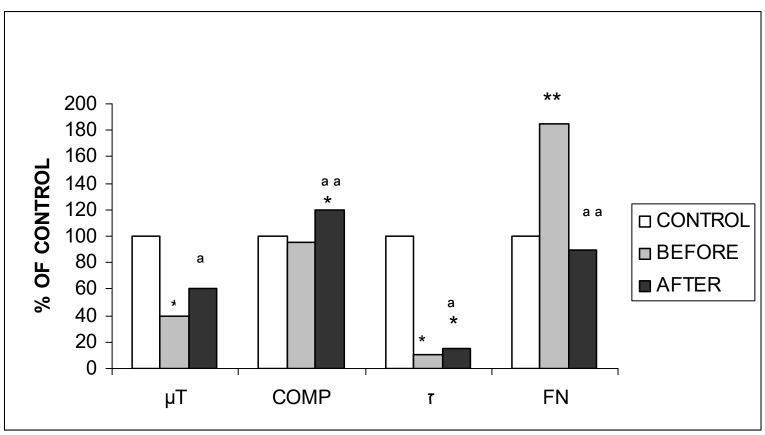

Fig. 2. Alteration in Fibrin Network Characteristics by Bezafibrate
*$p<0.05$, **$p<0.001$ when diabetic patients before and after treatment are compared with control group
[a]$p< 0.05$, [aa]$p< 0.001$ when diabetic patients after treatment with bezafibrate are compared with before treatment

administration of bezafibrate. The fibrin fiber thickness (μ_T) and permeability of the networks were significantly lower than showed significant increase ($p<0.05$). Permeability or the Darcy's constant also increased significantly ($p<0.05$). There was a decrease in the tensile strength of the networks depicted by an increase in compaction values ($p<0.001$). Plasma fibrinogen concentration decreased significantly ($p<0.001$).

Alterations induced by bezafibrate seemed to correct the parameters studied by bringing about the changes in the direction of control values.

DISCUSSION

Theories regarding the aetiology of atheroma implicate an injury response mechanism triggered by initial damage to the intima of the arterial vessels. The simultaneous presence of high thromboplastin activity and little or no fibrinolytic activity found in the intima and media suggest that fibrin can easily be deposited after tissue injury (Astrup *et al.*, 1959). Fibrin peptides, which are released during fibrin formation by the action of thrombin, may further contribute to pathogenesis by enhancing chemotoxins, of inflammatory cells, vasoconstriction and induction of cell proliferation (Cook and Ubben, 1990).

The results show a significant decrease in the plasma fibrinogen levels along with a significant decrease in the plasma lipid level and blood glucose levels. Similar changes have been reported previously (Mikhailidis *et al.*, 1990; Jeck *et al.*, 1997; Meade, 2001). However, the study on network characteristics present very interesting picture. The thickness of the fibers, μ_T, has significantly increased. This could be due to enhanced lateral polymerization or reduced end to end polymerization. It has been previously known that plasma fibrinogen levels have a negative correlation with the fibre diameter (Dhall and Nair, 1990). There are also significant increases in the compaction and permeability of the network. The increased compaction denotes a decrease in tensile strength. This may result from the fact that the fibrin in the network is less crosslinked. Such networks are more susceptible to lysis (Nair *et al.*, 1989).

The results described in this study indicate that the fibrin fibre size is not only the determinant of the fibrinolytic potential of the network. Other characteristics like degree of cross linking may also play an important role in determining this property. Fibrin monomer concentration in the protofibril network may also induce changes in the final characteristics of the network. Furthermore, certain agents may interact with plasminogen binding of various lipids and protein such apoprotein B and Lp (a).

It is now believed that plaques in the arteries arise from a stimulation of cellular in growths in response to deposited fibrin (Thomson and Smith, 1989; Ibid, 1994). Fibrin appears to be constantly deposited and lysed in proliferate atherosclerotic lesions, providing a continuing source of fibrin degradation products (FDP). FDP thus provide a continuing proliferate stimulus to smooth muscle cells (SMC) in the atherosclerotic lesion. FDP may also be chemotactic to leukocytes including macrophages. If the rate of deposition of fibrin exceeds its rate of lysis, fibrin will accumulate in the lesions. Speculatively, it may increase the SMC proliferation by providing a scaffold along which cells migrate. There is evidence that it binds lipoprotein, thereby contributing to the accumulation of lipid in lesions (Smith and Thompson, 1992; Smith *et al.*, 1994). This subendothelial fibrin deposition is composed of the fibres which are tightly packed, have higher tensile strength and are generally resistant to fibrinolysis. This intimal deposition of thin matrix may be because of higher circulating fibrinogen. Previous studies suggest that high concentration of fibrinogen in atherosclerosis-related intimal lesions indicates that there is binding within the intima of fibrinogen to LDL-cholesterol (Smith and Staples, 1981). Thus, plasma fibrinogen, fibrin and LDL-cholesterol will have a combined effect in promoting coronary atherosclerosis.

CONCLUSION

The current study substantiates the potential relationship of hypercholesterolaemia and thrombosis in the development of atherosclerosis in diabetic patients. It is clearly shown that lipid and fibrinogen lowering agent benzafibrate can also causes the changes in the plasma fibrin network structure of patients. The changes induced by the drugs, such as bezafibrate, in the plasma fibrin network characteristics along with the known effects of these agents on lipid make them the drug of choice for their additional benefits.

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